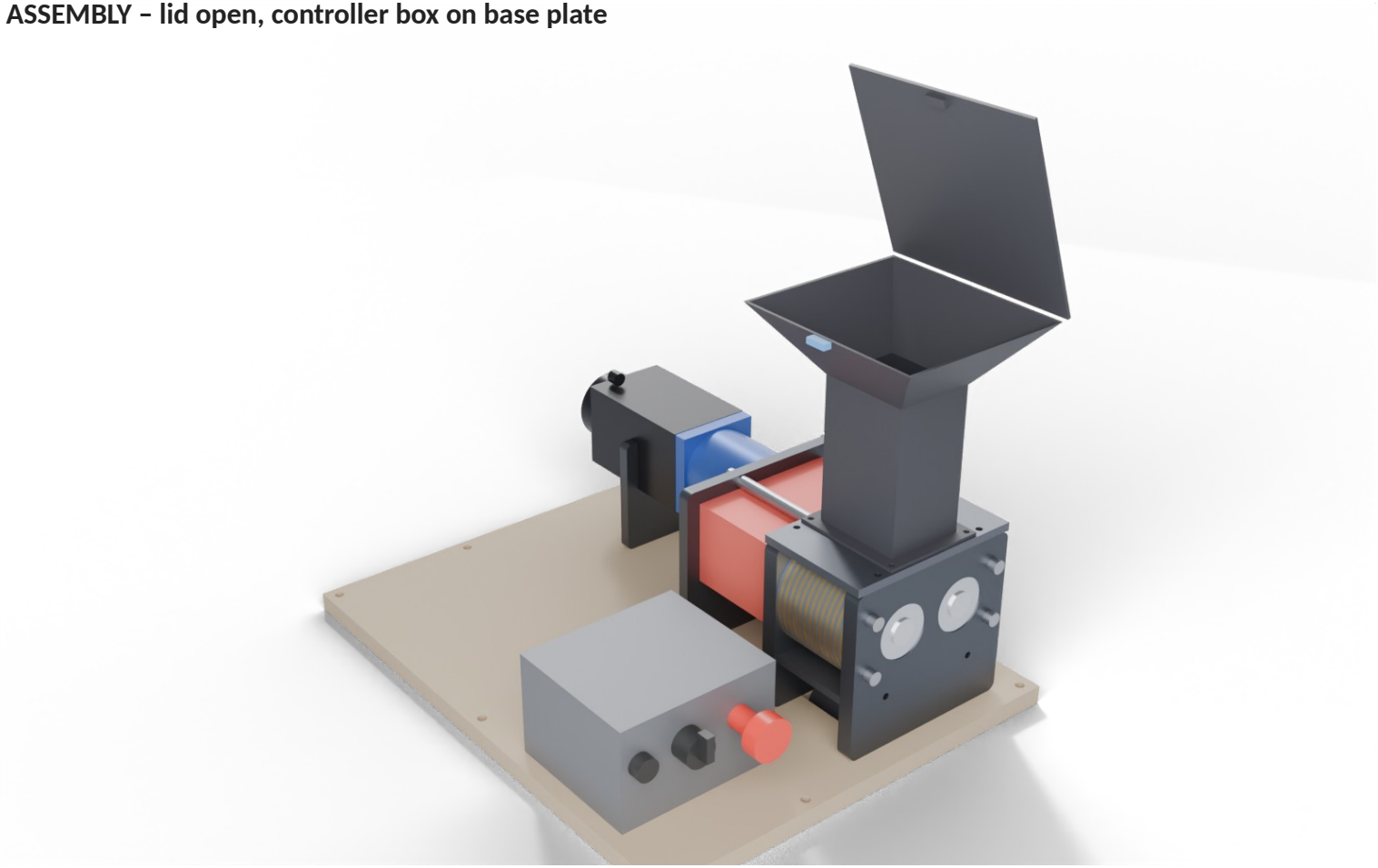
DIY DUAL-SHAFT SHREDDER – BLUEPRINT SET

ASSEMBLY – lid open, controller box on base plate



Models: shredder_assembly.fbx (Blender) · Cut files: shredder_dxf.zip

Dimensions marked VERIFY come from listings or photos, not drawings – see sheet 16.

26 + 26
blades + spacers (Ø76 / Ø58)

18 teeth
hooked, per blade – fine strips, not chunks

68 mm
shaft spacing; 1:1 gears sync the shafts

70 Nm
jam torque cap (driver set to 3.7 A); gearbox 45 Nm nominal

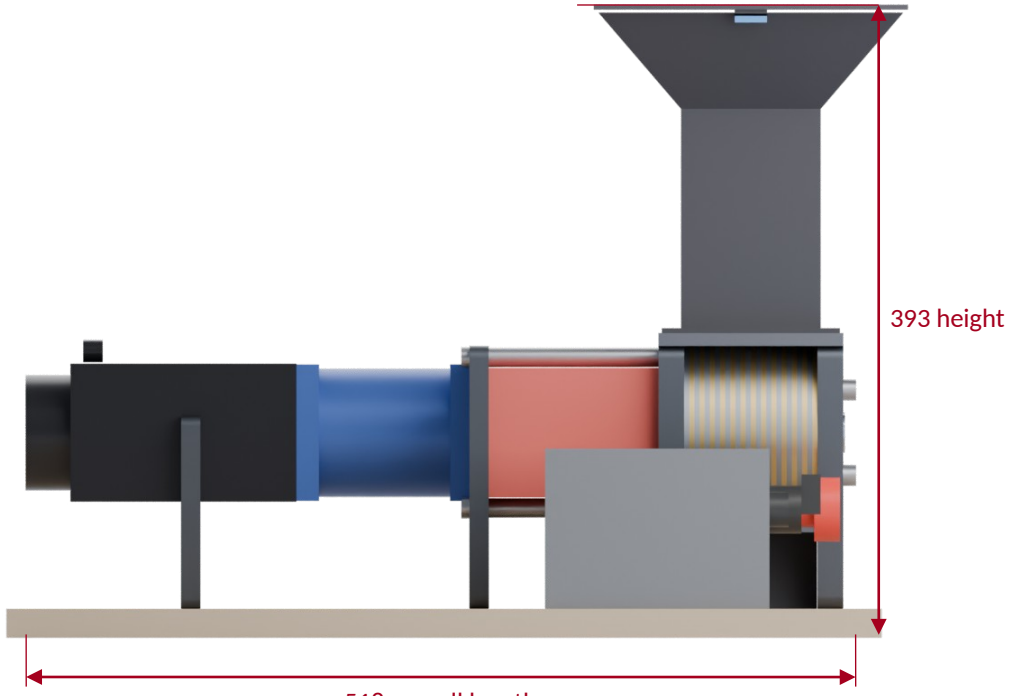
SHEET INDEX

- 1 Overview
- 2 Assembly views + overall size
- 3 Exploded view + part numbers
- 4 Cutter blade + spacer
- 5 Cutter stack (section)
- 6 Comb, wall + top plate
- 7 End plates (far / drive / motor)
- 8 Gear sync + drive chain
- 9 Drive-train stack-up
- 10 Motor cradle + plywood base
- 11 Calculations 1: torque, speed, cutting
- 12 Calculations 2: gears, bearings, mass
- 13 Safety + electrical
- 14 Bill of materials
- 15 Print / cut / buy
- 16 Assembly order + checks

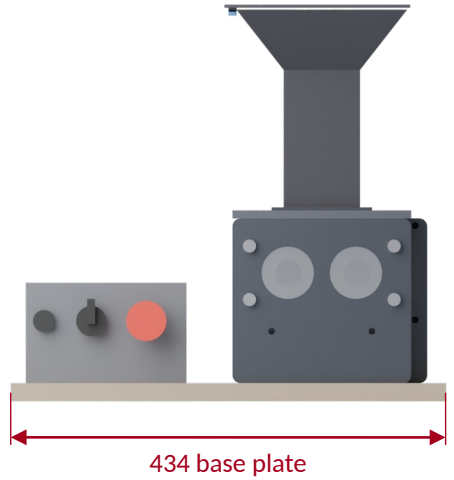
All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

ASSEMBLY VIEWS – OVERALL SIZE

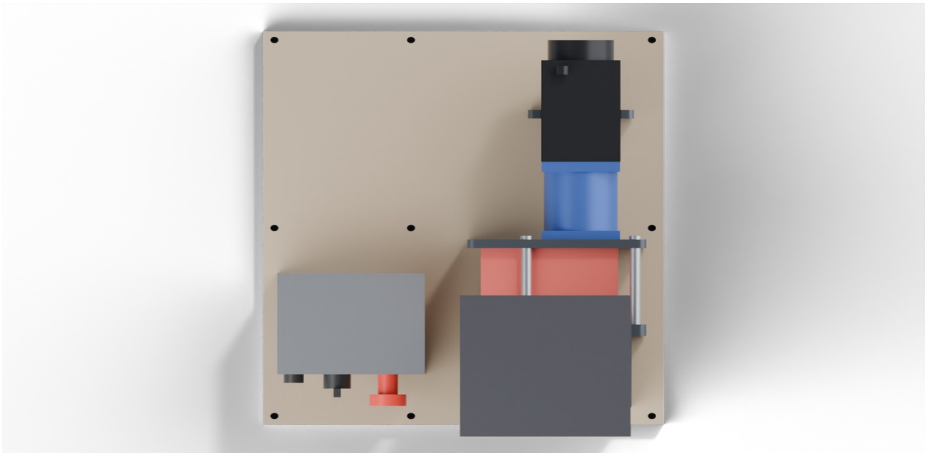
SIDE (from +X) – lid closed



FRONT (far end)



TOP – 434 × 546 base plate

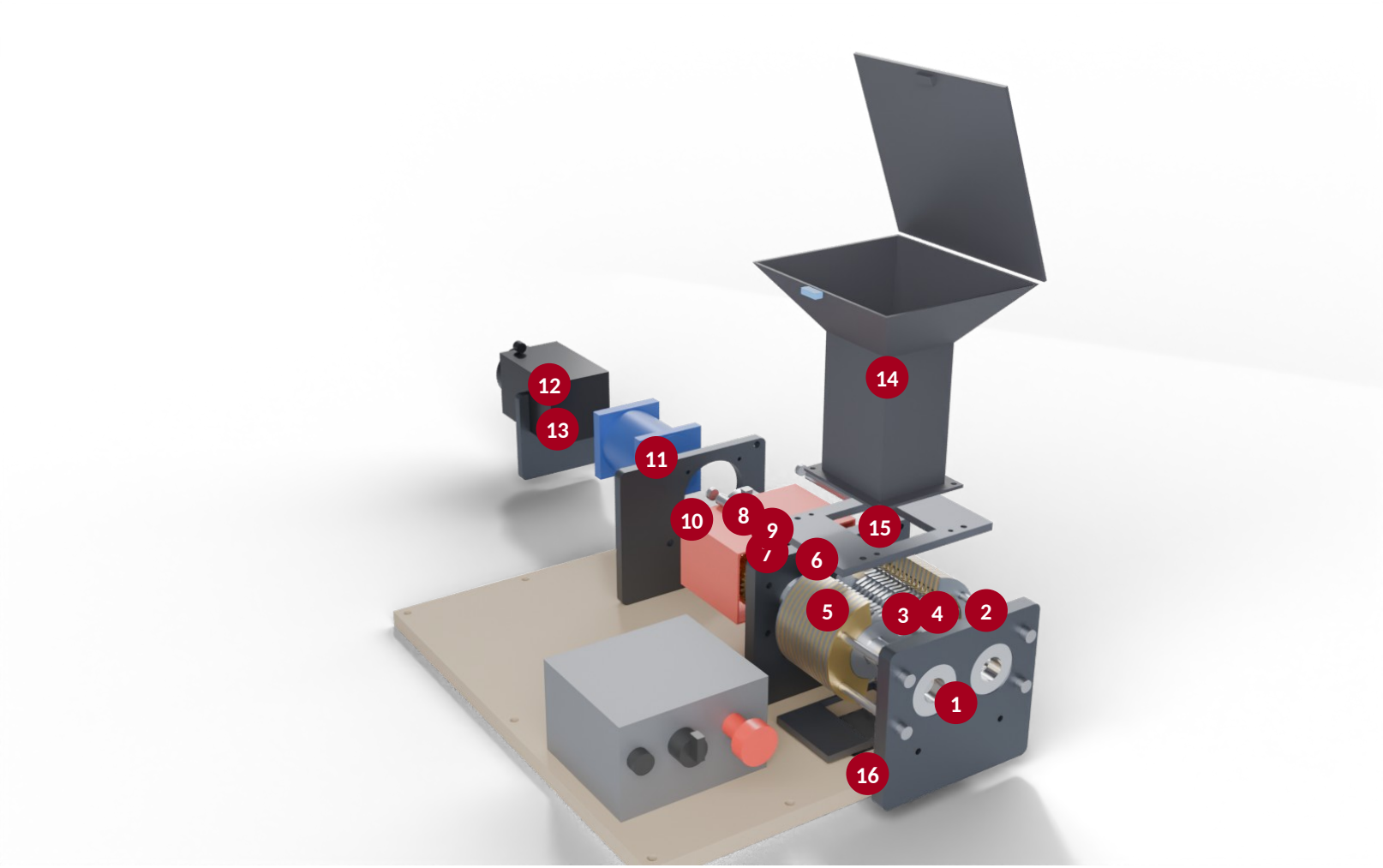


DESIGN NOTES

- Base plate 434 × 546 carries the machine, the motor cradle and the controller box; a cut-out lets the discharge chute drop through.
- Funnel overhangs the end plates by ≈ 33 mm each way – the 140 mm throat keeps hands away from the blades.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

EXPLODED VIEW – PART NUMBERS



Exploded along the shaft axis; the hopper and top / bottom plates are also lifted / dropped for clarity.

PARTS

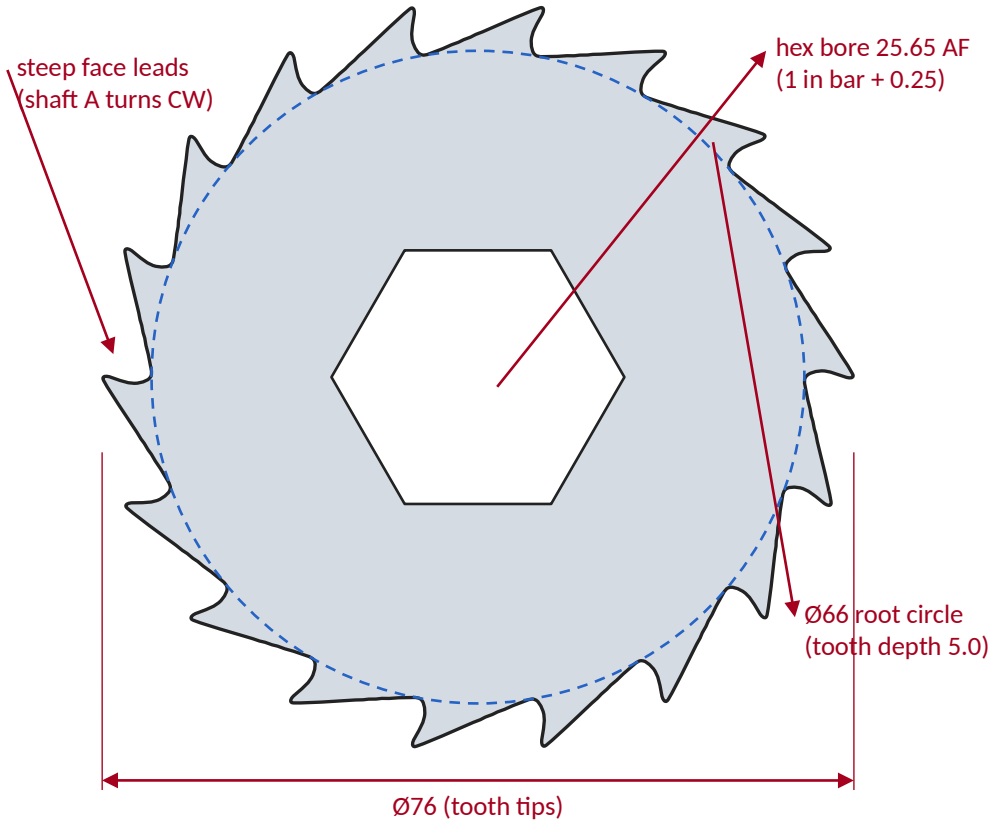
1 Far end plate (16 mm)
2 Tie rods M8 × 110 (4)
3 Cutter stack: 26 blades, 26 spacers
4 Combs + wall plates (52)
5 Drive end plate (16 mm)
6 Sync gears m2 × 34 T × 20 (2)
7 Gear guard (printed PETG)
8 Standoffs Ø12 × 106 (4)
9 XL2 jaw coupler D54 × 78
10 Motor mount plate (12 mm)
11 10:1 planetary gearbox
12 NEMA 34 closed-loop motor
13 Motor cradle (12 mm)
14 Throat + funnel + lid
15 Top plate (inlet 70 × 81)
16 Discharge chute

Not shown: bearings (4 × 6205), M6/M8 hardware, plywood base, controller box.

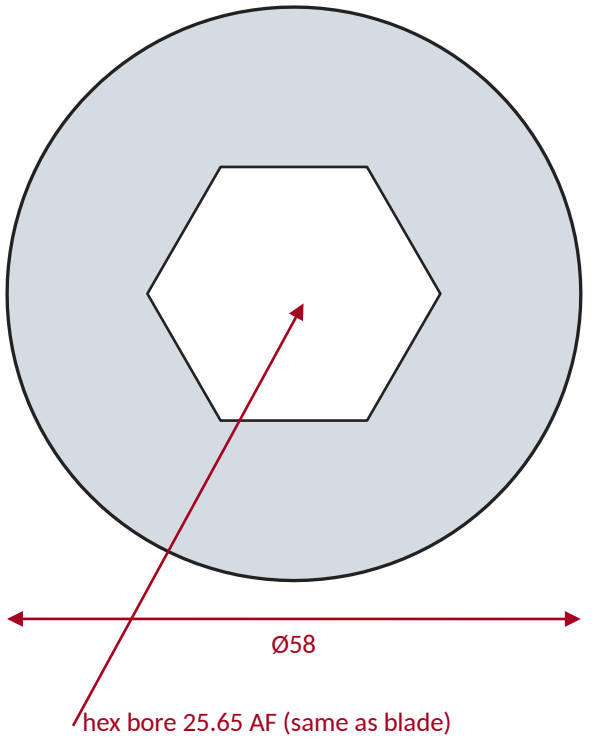
All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

CUTTER BLADE + SPACER – 3.0 AND 3.4 MM STEEL

BLADE (×26, 3.0 mm) – shaft A orientation



SPACER (×26, 3.4 mm)



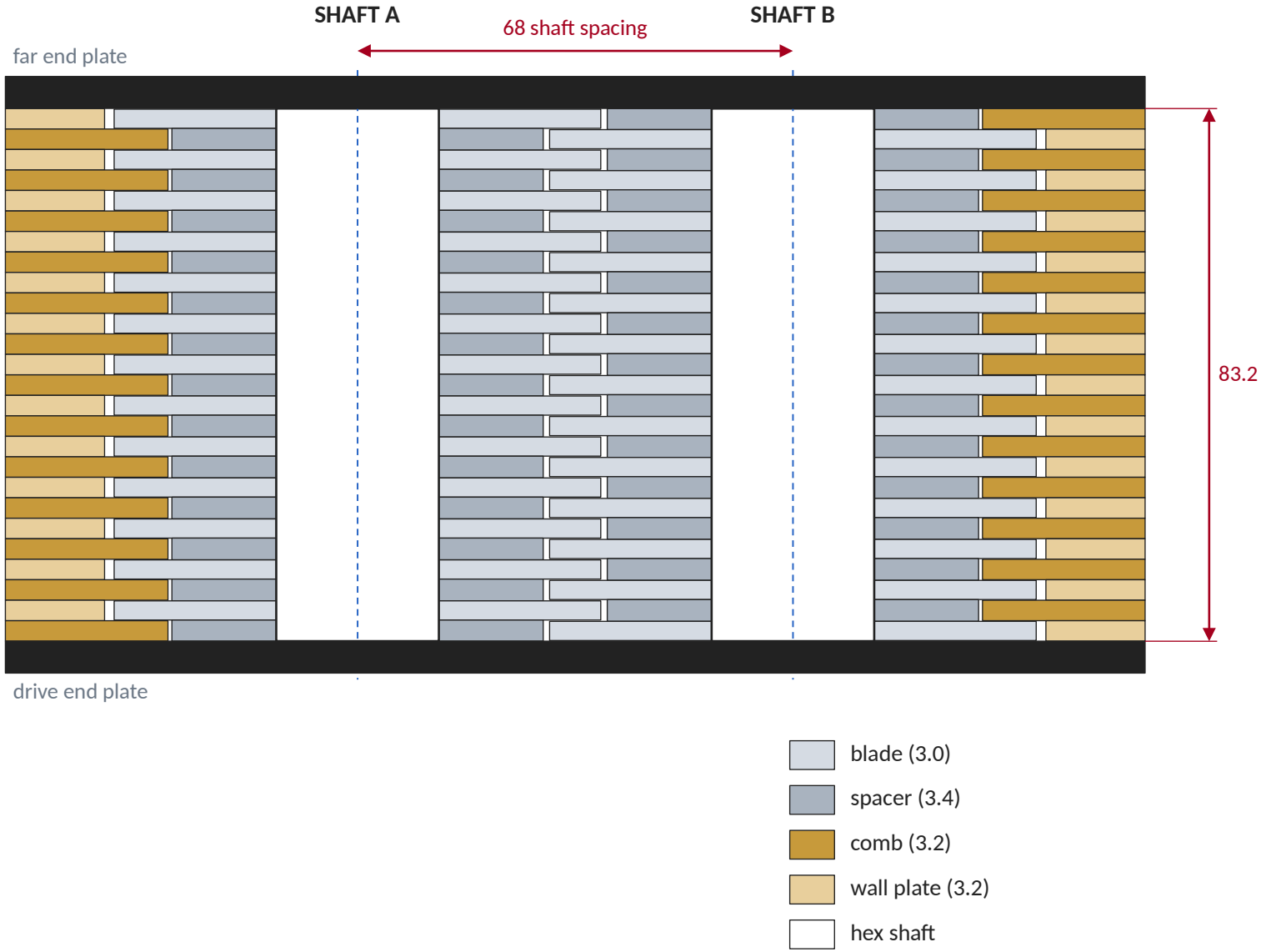
Blade tip clears the opposing spacer by 1.08 mm

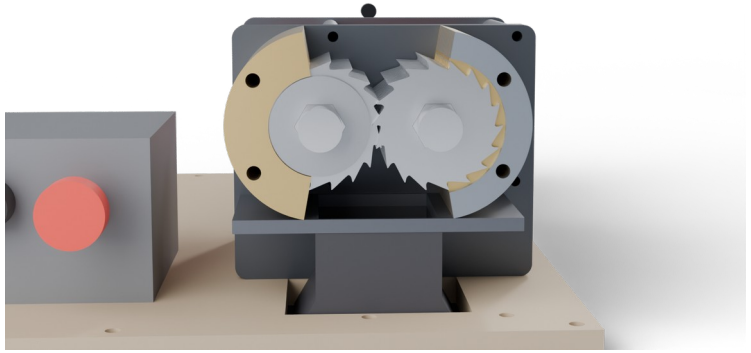
DESIGN NOTES

- Hook shape is traced from your Fusion "Shredi V2" blade and compressed to 18 fine teeth.
- Tooth pitch at the tips ≈ 13.3 mm; each blade shears a 3.2 mm wide strip (estimate of bit size).
- One blade file: cut 26, then turn 13 over so the hooks mirror on shaft B.
- Cheapest start: mild steel plate. Move to 3 mm AR400 / AR500 only if the teeth dull on abrasive scrap.
- Hex bore = 1 in hex bar (25.4 AF) + 0.25 mm slip fit.
- Spacers are 0.4 mm thicker than blades – this leaves 0.1 mm each side for the stationary plates.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

CUTTER STACK – SECTION THROUGH BOTH SHAFT AXES





Blade hooks meshing – view from the far end

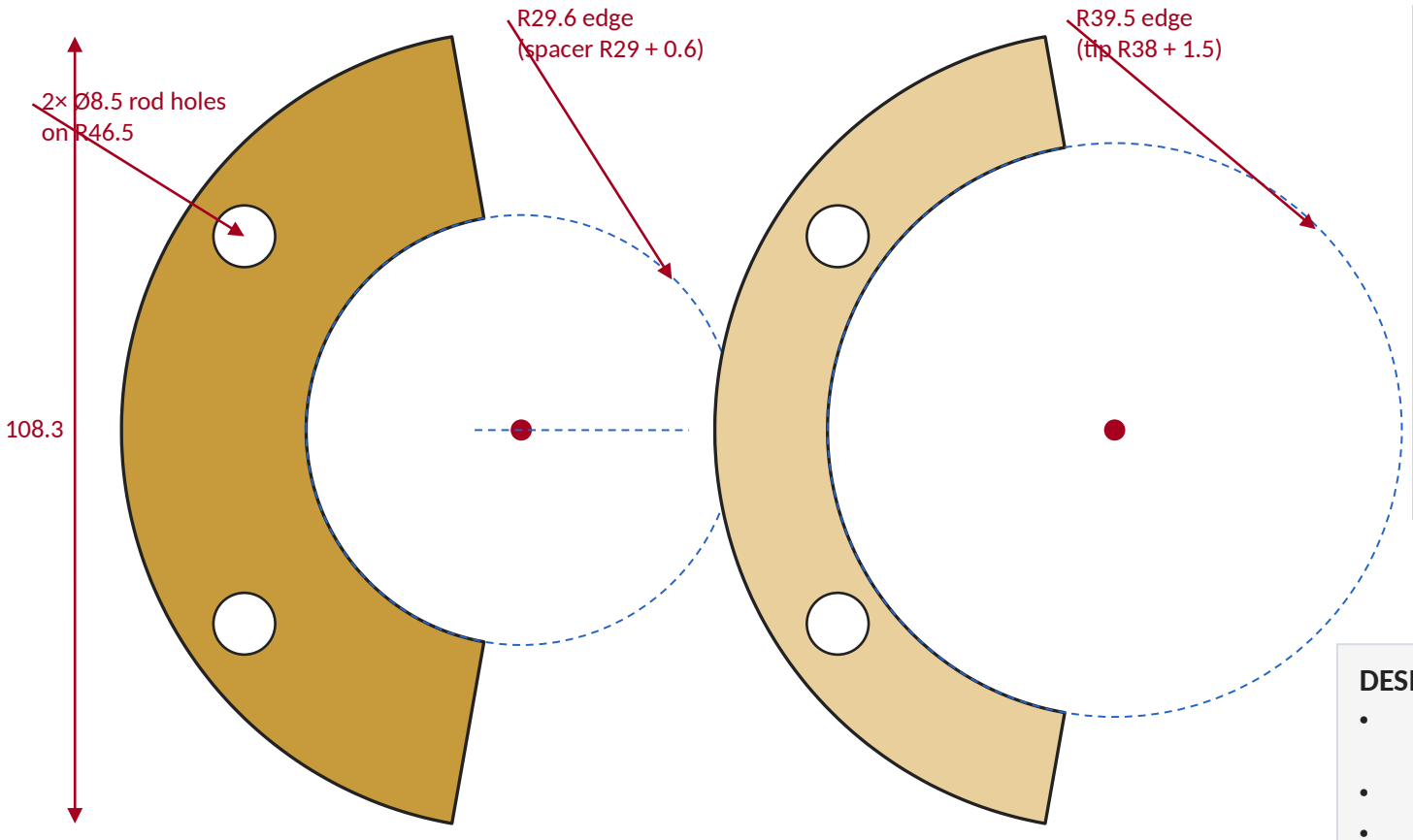
DESIGN NOTES

- Every layer is 3.2 mm: A and B blades alternate, 13 each = 26 layers = 83.2 mm.
- Each blade sits in the other shaft's spacer layer and shears against its neighbours (0.2 mm axial).
- Stationary combs touch the spacer zone; wall plates close the blade layers.
- Gaps: 0.1 mm blade ↔ plate, 1.0 mm tip ↔ spacer, 1.5 mm tip ↔ wall.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

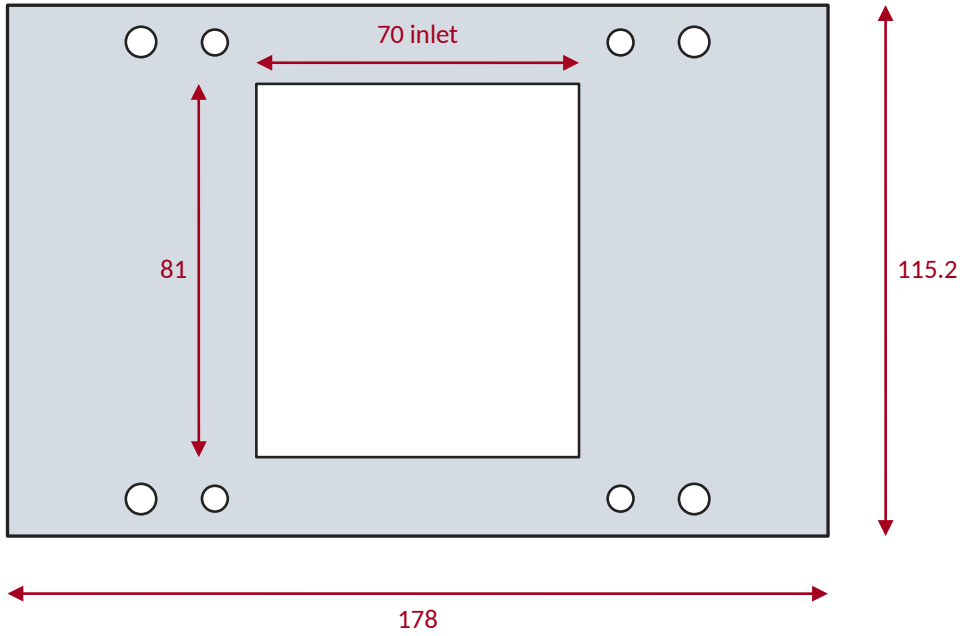
STATIONARY PLATES – COMB, WALL AND TOP PLATE

COMB (×26, 3.2 mm)



WALL (×26, 3.2 mm)

TOP PLATE (×1, 8 mm)



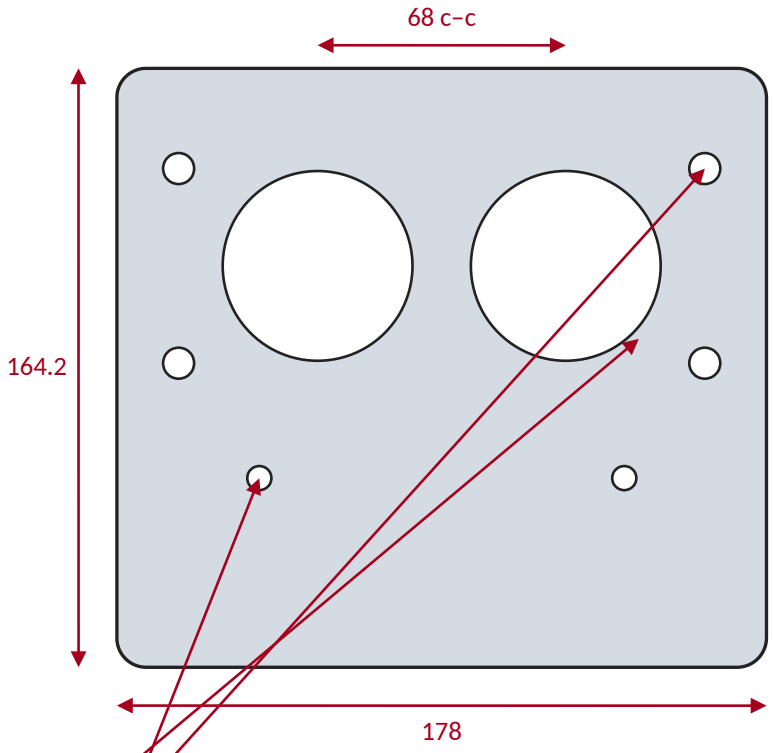
DESIGN NOTES

- Comb and wall plates alternate: comb at the spacer layers, wall where the blades run – 26 of each.
- Plates stack flush on the four M8 tie rods (≥ 4 mm web at the rod holes).
- Top plate: slot 70 × 81 (spec inlet 8.1 × 7 cm), 4× Ø5.5 for the throat flange, 4× Ø6.6 into the end-plate top edges.
- Bottom plate is the same size with a 70 × 78.4 discharge slot (DXF 10).

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

END PLATES – FAR (16 MM), DRIVE (16 MM), MOTOR MOUNT (12 MM)

FAR END PLATE (×1)

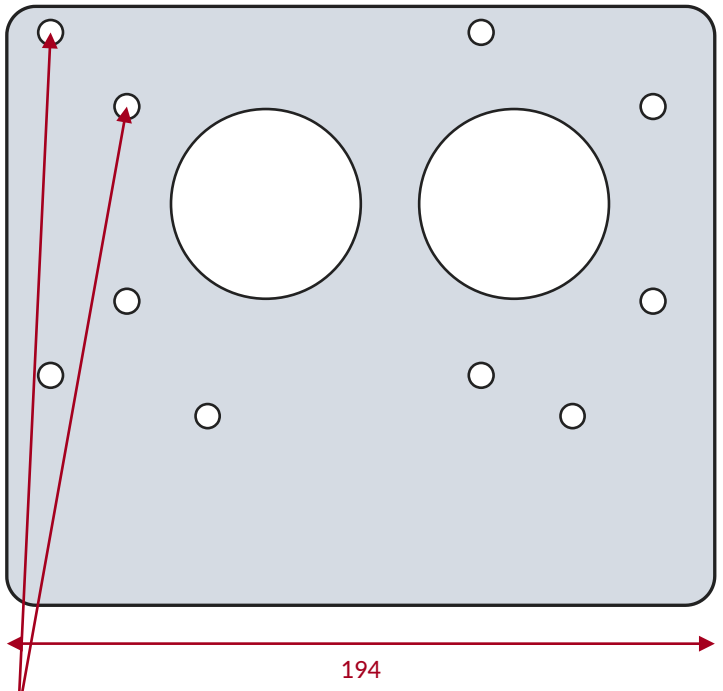


2× Ø52.0 bearing bores (6205)

4× Ø8.5 tie-rod holes

2× Ø6.6 bottom-plate screws

DRIVE END PLATE (×1)

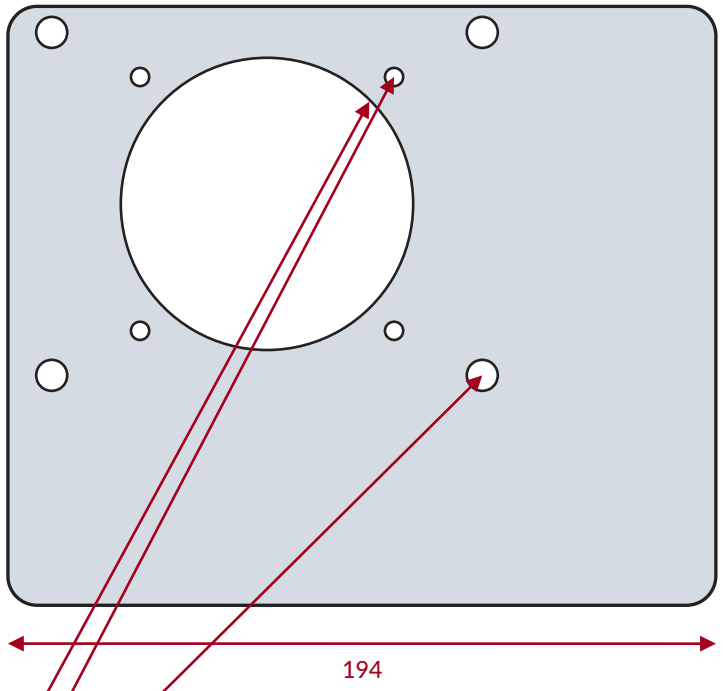


4× Ø6.8 = M8 tap (rods thread in)

4× Ø6.8 M8 tap (standoffs)

Bores + rod pattern as the far plate

MOTOR MOUNT PLATE (×1)



Ø80.1 gearbox pilot seat (80 H7)

4× M6 tap (Ø5.0 drill) on 69.6 sq

4× Ø8.5 standoff screws

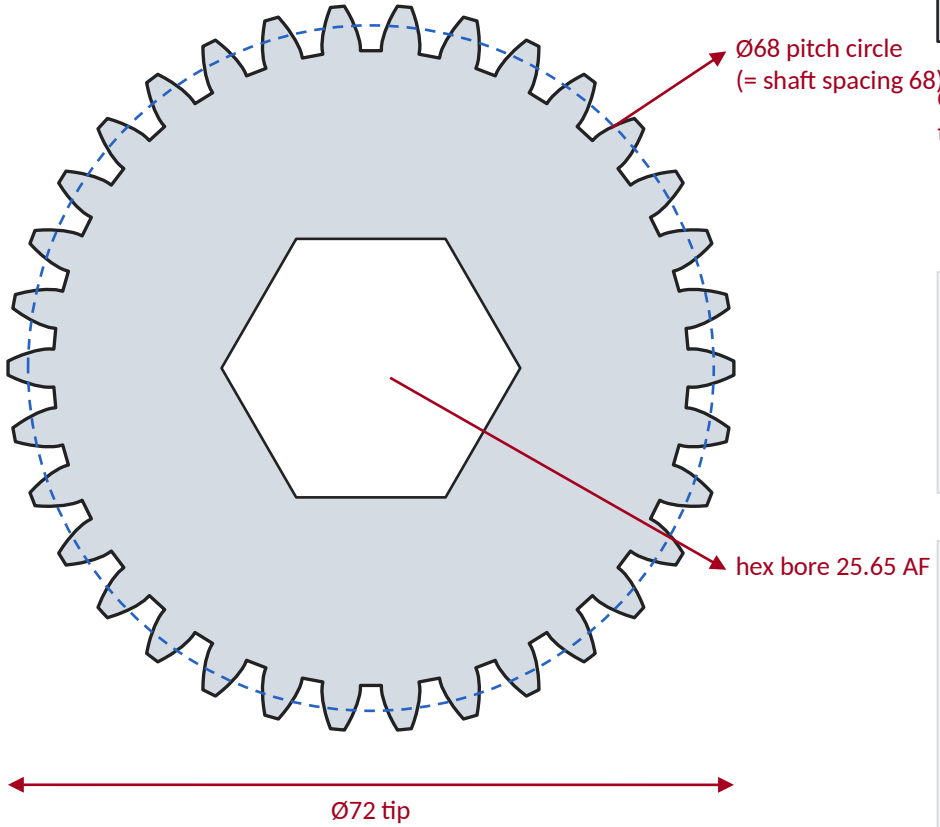
NOTES

- Press bearings into the Ø52 bores (laser bores run slightly loose: retain with bearing compound). Bottom edges get 2× M8 taps per plate for the base plate. Motor plate: gearbox bolts (M6 × 20) go in from the gearbox side.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

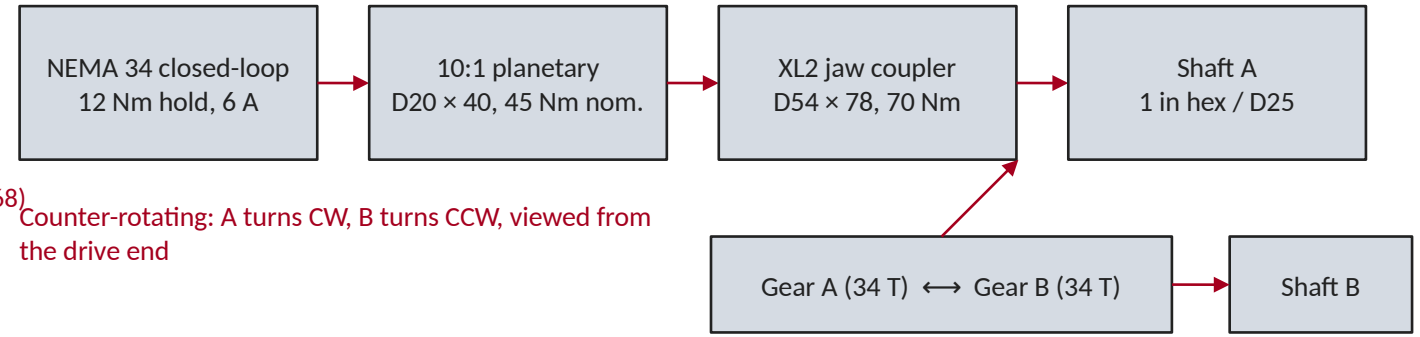
GEAR SYNC PAIR + DRIVE TRAIN

SYNC GEAR (×2, 20 mm) – m2, 34 T, 20°



Mesh check: no overlap, 0.09 mm backlash at 68.0 centres

POWER PATH



Counter-rotating: A turns CW, B turns CCW, viewed from the drive end

70 Nm
jam torque cap at 3.7 A driver current (10 × 0.95 × 7.4 Nm)

45 Nm
gearbox nominal; the continuous working limit

1.85 kN
blade-tip force at the cap

60 rpm
at 600 rpm motor; tip speed 0.239 m/s

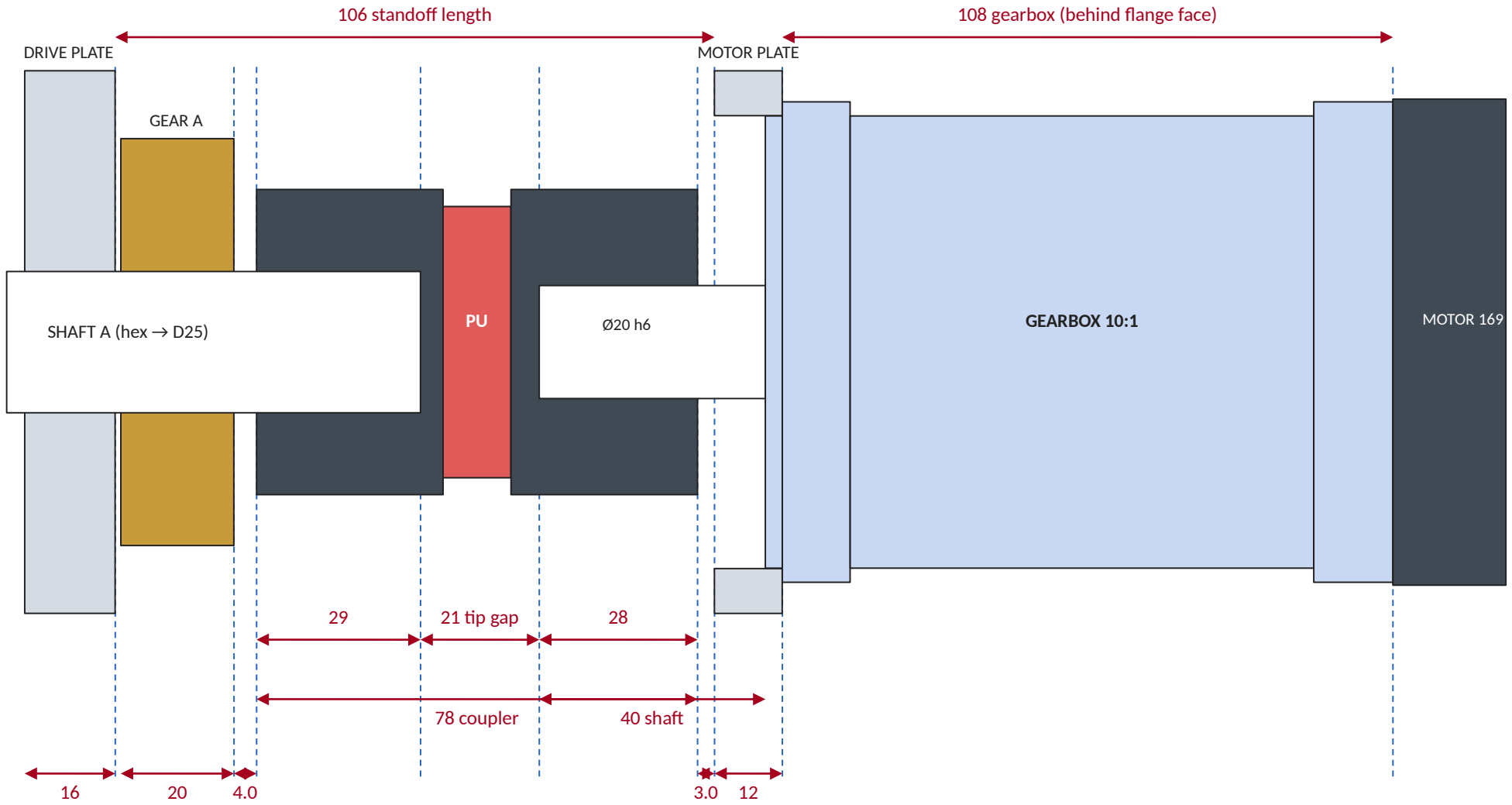
DESIGN NOTES

- A 1:1 gear pair is the “part for counter-rotation”; the second shaft takes no motor of its own, so one gearbox is enough.
- Gears: buy pre-hardened 4140 plate (≈ 300 HB as bought, no heat treat) and have it waterjet cut. Mild steel fails the contact-stress check (sheet 12).
- Without a current limit the motor stalls at 114 Nm – above the gearbox peak (90) and the coupler (70). Set the driver peak to 3.7 A.
- Torque-vs-speed fractions are assumptions – check the motor curve at 60 V.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

DRIVE-TRAIN STACK-UP – GEAR TO MOTOR

SECTION THROUGH SHAFT A – w measured from the far end plate (mm)



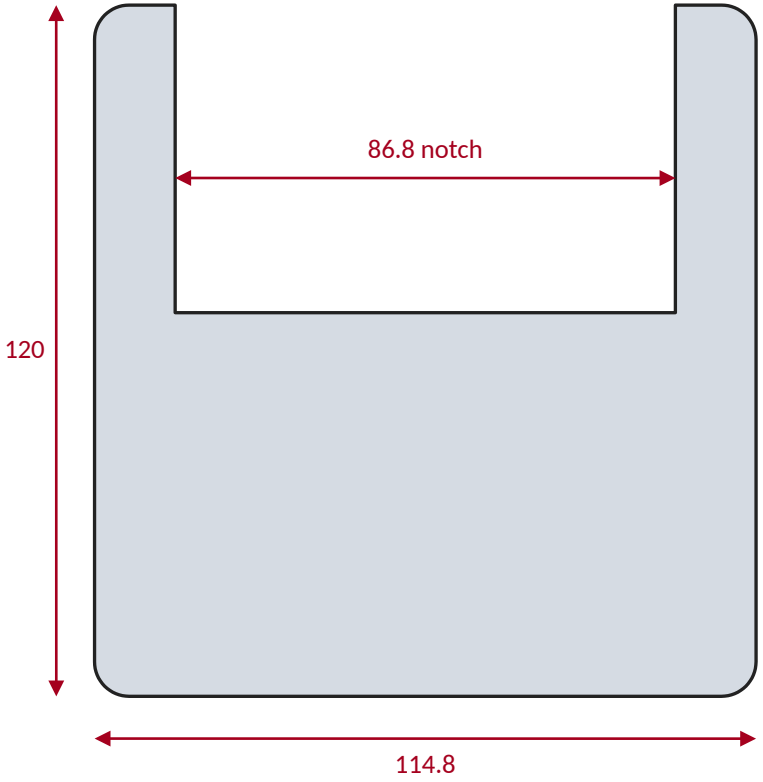
STACK-UP

- 106 = 1.0 + 20 (gear) + 4.0 + 78 (coupler) + 3.0.
- Shaft A ends 1.0 short of the hub bore bottom (29 engaged); the gearbox shaft is 28 engaged, 2.0 short.
- Shaft tips sit 21 apart inside the 18 mm spider zone.
- Gearbox shaft length is taken from the pilot tip. If the seller means the flange face, engagement drops to 25 mm.
- Clamp heads clear the gear guard by 7 mm.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

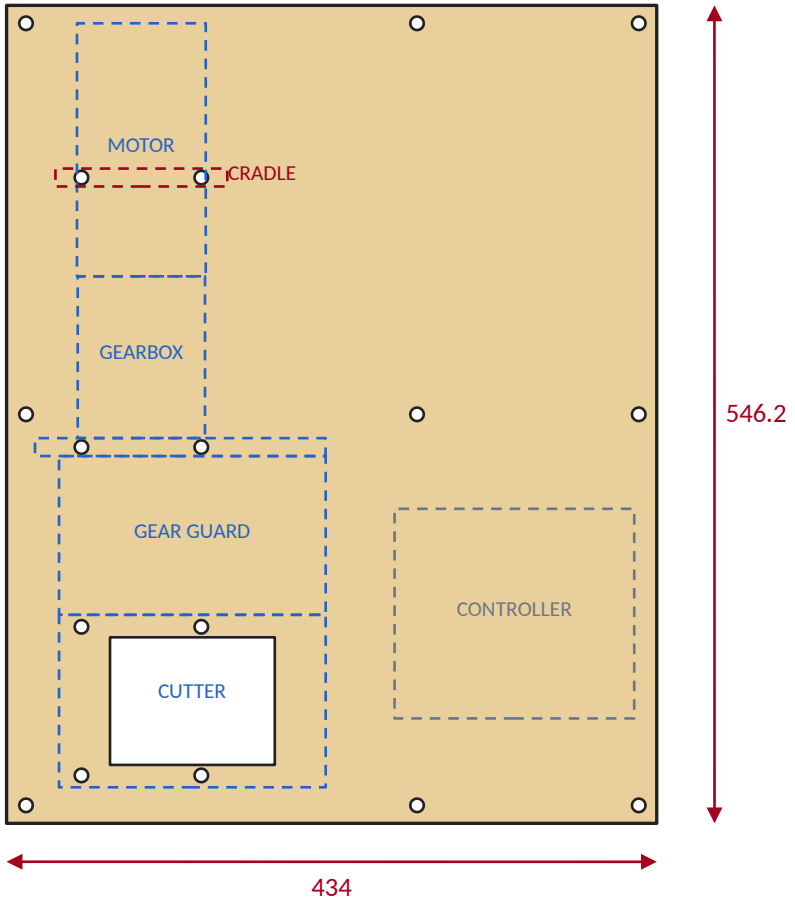
MOTOR CRADLE + PLYWOOD BASE

MOTOR CRADLE (×1, 12 mm)



Notch floor sits 0.4 mm below the motor underside; the motor drops in from the top. Two M8 screws go through the base into the bottom edge (u = −74, +6).

BASE (×1, 18 mm plywood) – plan, drive end at the bottom



NOTES

- Plywood base (18 mm): 2.77 kg; whole machine ≈ 36.8 kg (the 10 mm steel base was 18.6 kg).
- The base is the one big part you cut yourself: print the DXF at 1:1 (tiled) or use it as a CNC file.
- Centre of gravity: u 13.7, w 158.2, 90.9 mm above the base – 137.7 mm inside the nearest edge (static tip angle 56.6°).
- Motor + gearbox (7.9 kg) hang up to 280 mm off the motor plate = 11.7 N·m. The cradle carries the weight; the gearbox bolts only see 84 N each.
- Dashed outlines: blue = machine, red = cradle, gray = controller box.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

CALCULATIONS 1 – TORQUE CHAIN, SPEED, POWER, CUTTING

TORQUE CHAIN

Step	Value	Basis
Motor holding torque	12.0 Nm	6.0 A → 2.0 Nm / A
Gearbox ratio × efficiency	10 × 0.95 = 9.5	listing: ≥ 95 %
Gearbox nominal / peak	45 / 90 Nm	listing
Coupler nominal	70 Nm	XL2 table
Weakest link on a jam	70 Nm (coupler)	min of the above
Motor torque for 70 Nm	7.37 Nm	70 ÷ 9.5
Driver peak current (set)	**3.7 A = 62 % of 6.0 A**	**→ 70.3 Nm out**
Current for 45 Nm continuous	2.37 A	45 ÷ 9.5 ÷ 2.0
Stall with no limit	114 Nm	exceeds gearbox peak!

SPEED POINTS (motor torque fraction is assumed)

Motor rpm	Shaft rpm	Torque avail.	Out Nm	Out W	Tip m/s
300	30	75 % of hold	70.3	221	0.119
600	60	60 % of hold	68.4	430	0.239
1000	100	40 % of hold	45.6	478	0.398

Supply: 60 V × 10 A = 600 W; × 55 % stepper-system efficiency (assumed) × 0.95 gearbox ≈ 314 W at the shaft = 50 Nm at 60 rpm.

Pulse rate for the Nano at 1600 pulses/rev: 16 kHz at 600 rpm, 26.7 kHz at 1000 rpm – well inside the driver limit.

CUTTING: tooth force = T ÷ R_blade = 1184 N at 45 Nm, 1850 N at the cap

Material (thickness)	Shear τ MPa	Force to shear (τ × t × 3 mm)	Margin at 45 Nm
PET bottle wall 0.3 mm	50	45 N	26.3 ×
HDPE 2 mm	22	132 N	9 ×
PLA print 4 mm	45	540 N	2.2 ×
ABS 5 mm	35	525 N	2.3 ×
Acrylic 6 mm	55	990 N	1.2 ×

WHAT TO SET / ASSUME

- CL86T: peak current 3.7 A (jam torque ≈ 70 Nm); continuous draw follows the load.
- One tooth can shear PLA up to ≈ 8.8 mm thick, HDPE ≈ 17.9 mm at 45 Nm.
- Shear strengths are handbook values; thick prints will stall long before the bottles do.
- Torque fractions at 300 / 600 / 1000 rpm are assumed for an 86 mm closed-loop stepper at 60 V – check the motor curve.

CALCULATIONS 2 – GEARS, SHAFTS, BEARINGS, MOUNTING

GEAR MESH (Lewis Y = 0.371, Kv = 1.035; allowable: 4140 HT 248 / 840 MPa, A36 133 / 336 MPa)

Load case	Face mm	Tooth force N	Bending MPa	Contact MPa	4140 HT	A36
nominal 45 Nm	12	1324	154	855	over	fails
nominal 45 Nm	20	1324	92	662	OK	fails
jam at current cap 70 Nm	12	2068	240	1069	over	fails
jam at current cap 70 Nm	20	2068	144	828	OK, tight	fails
unlimited peak 90 Nm	12	2647	308	1209	over	fails
unlimited peak 90 Nm	20	2647	185	936	over	fails

Mesh: no overlap, 0.0947 mm clearance, contact ratio ≈ 1.68. Chosen: 20 mm face, hardened, driver cap 3.7 A.

SHAFTS, BEARINGS, MOUNTING

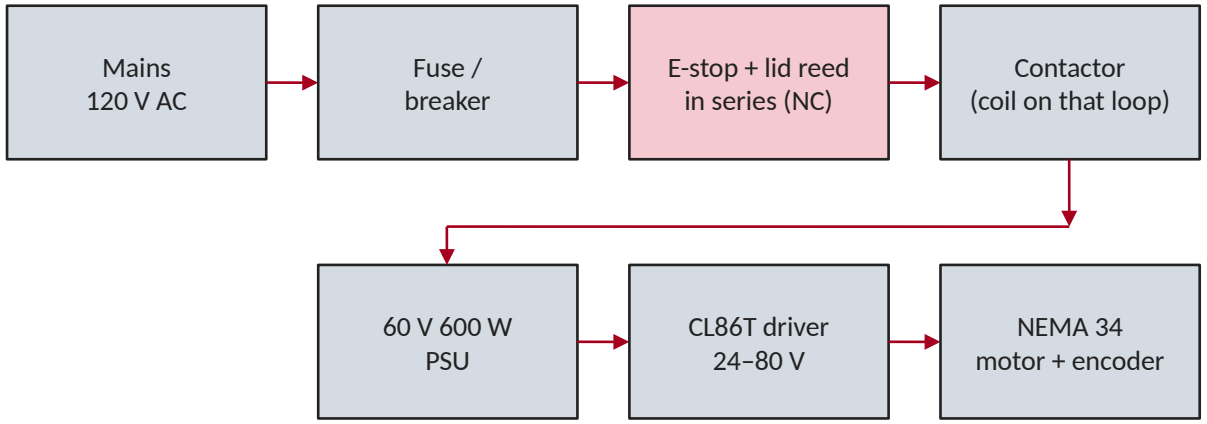
Check	Result	Limit / note
Torsion, D25 seat at 70 Nm	22.9 MPa	allow ≈ 202 MPa (1018) → SF 8.8
Hex drive, whole torque on 1 blade	89 MPa	AR400 ≥ 1000, mild ≈ 250
Drive bearing (gear + blade force)	4487 N	6205 static C0 7800 N → ×1.7
Gearbox flange bolts (M6 × 20)	84 N tension	357 N shear from torque reaction
Motor shaft in gearbox bore	37 in 41.5	D14 F7 clamp, remove the key

MASS + BALANCE

- Total ≈ 36.8 kg: base 2.77, plates 9.86, motor 5.5, controller 3, gearbox 2.4.
- Centre of gravity 90.9 mm above the base; static tip angle 56.6°.
- Motor mass (5.5 kg) and the controller box (3 kg) are estimates.
- The gear pair is the part to watch: if shaft B jams, all of the torque goes through the mesh.
- Stresses are hand calculations (Lewis / Hertz, no life factor) – a screening check, not a certified design.

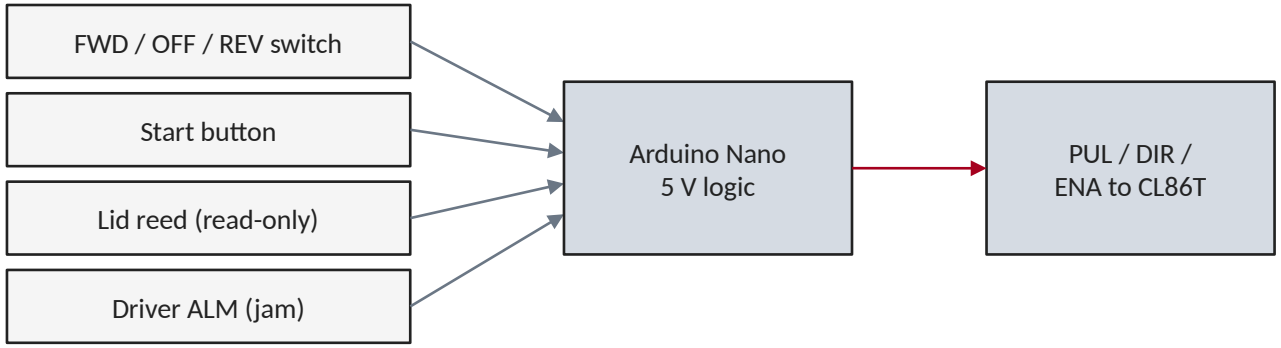
SAFETY CHAIN + ELECTRICAL BLOCK DIAGRAM

POWER + E-STOP (hardware path)

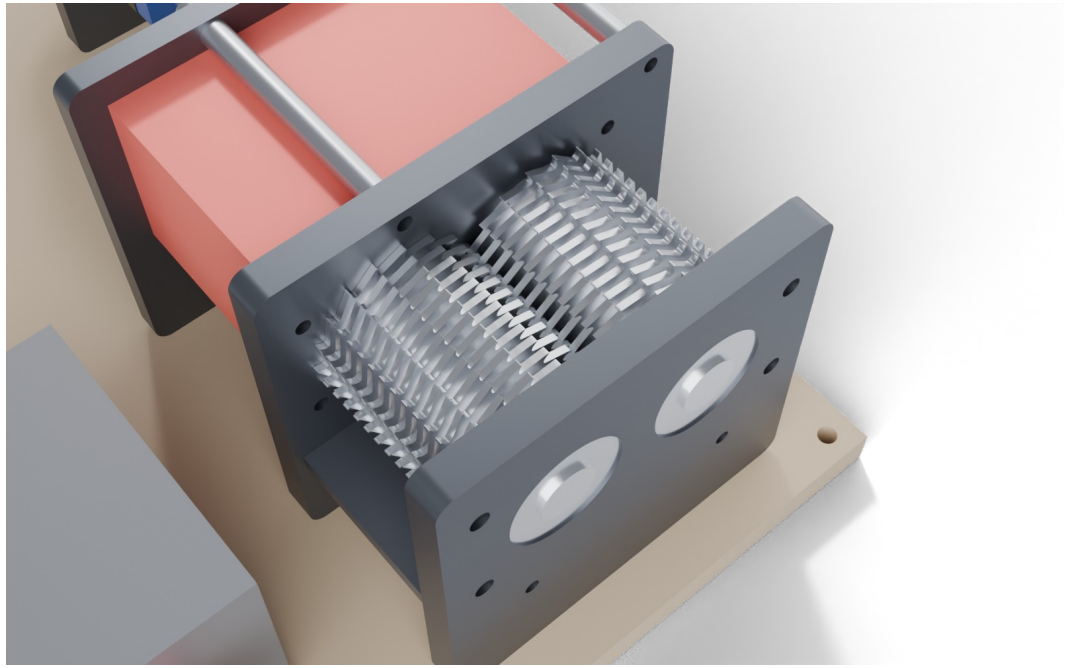


Contactor drops the PSU feed on E-stop or lid open – no software in the loop.

CONTROL (logic)



Nano auto-reverses on a jam (ALM) and holds ENA low whenever the interlock is open.



Cutter chamber with the top plate, hopper and side walls hidden

GUARDS + SAFETY NOTES

- Throat 70 × 81 × 140 mm tall plus a hinged lid with a reed-switch interlock.
- Red gear guard (printed PETG, 2.4 mm walls) closes the gear zone between the plates; E-stop is on the controller front.
- Set the CL86T peak current to 3.7 A: it caps the jam torque at the coupler rating (70 Nm).
- Counter-rotating steel blades pull in anything – never reach into the throat; always stop and wait.
- Have mains wiring checked by a qualified person; this is not a certified guard design.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

BILL OF MATERIALS

#	Part	Qty	Size / material	Source
1	Blade	26	Ø76 × 3.0 mm steel, 18 teeth	DXF 01 (turn 13 over)
2	Spacer	26	Ø58 × 3.4 mm steel	DXF 02
3	Comb plate	26	3.2 mm steel	DXF 03
4	Wall plate	26	3.2 mm steel	DXF 04
5	Hex shaft	2	1 in hex bar; A = 170 mm, B = 139 mm; ends turned Ø25 h6	bar stock + one machine-shop job
6	Sync gear	2	m2 × 34 T, 20 mm, hex bore, pre-hard 4140 (≈ 300 HB)	DXF 05, waterjet, no heat treat
7	End plates + motor plate	3	16 / 16 / 12 mm steel	DXF 06–08
8	Top + bottom plate	2	8 mm steel	DXF 09–10
9	Base	1	434 × 546 × 18 mm plywood	DXF 11 as template, jigsaw
10	Motor cradle	1	12 mm steel U-saddle, 114.8 × 120	DXF 12
11	Bearing 6205-2RS	4	25 × 52 × 15	bearing supplier
12	Tie rod (M8 SHCS)	4	M8 × 110, threads into drive plate	hardware
13	Standoff + screw	4	Ø12 × 106 tube, M8 × 130 SHCS	hardware
14	Shaft coupler	1	Sinoblu XL2 D54 × L78, 1045 steel, bores 25 / 20, 70 Nm	listing $18.94 (selected)
15	10:1 planetary gearbox	1	YEJMKJ, 85 sq, Ø20 × 40 shaft, 45 / 90 Nm; + 4× M6 × 20 SHCS	listing $89.99 (selected)
16	NEMA 34 closed-loop kit	1	34HS59-6004D-E1000 12 Nm + CL86T V4.1 + cables	listing $155.00
17	60 V 10 A PSU	1	600 W, 110 V in	listing $31.99
18	Arduino Nano	1	3-pack, USB-C	listing $15.99
19	E-stop, contactor, switches, fuse	1 set	NC mushroom, reed switch, FWD/OFF/REV	electrical supplier
20	Hopper, throat, lid, chute	1 set	3D-printed (PETG) – models in the FBX	print
21	Gear guard	1	148.8 × 80.8 × 106 shell, 2.4 mm walls, PETG	print

Listed purchases: $311.91 (motor kit 155.00, gearbox 89.99, PSU 31.99, coupler 18.94, Nano 15.99). Gearbox and coupler are selected, not yet ordered.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

WHAT TO PRINT, WHAT TO CUT, WHAT TO BUY

WHO MAKES WHAT

Route	Parts	Why
3D print (PETG / ASA)	Hopper flange, throat guard, lid, discharge chute, gear guard, controller box	Carry no cutting or drive load; free and fast to change
Laser cut (mild steel)	Blades, spacers, combs, walls, 5 plates, motor cradle	110 flat parts: not practical by hand
Waterjet (4140 pre-hard)	Two sync gears, 20 mm	Hardened teeth with no heat treat; the only hard-to-source part
Plywood, jigsaw	Base 434 × 546 × 18 mm	Replaces the 18.6 kg steel base; 15.8 kg lighter
Machine shop, one job	Turn 4 hex-bar ends to Ø25 h6	Needs a lathe; the round coupler bore means a hex bearing cannot replace it
Buy	Motor kit, gearbox, coupler, PSU, Nano, 6205 bearings, hex bar, M6 / M8 hardware, E-stop set	Listed purchases are $311.91

CUT ORDER SUMMARY (for quotes)

Part	Qty	Thk	Size mm	Cut m
Blade	26	3.0	75.8 × 74.7	10.9
Spacer	26	3.4	58 × 58	7
Comb plate	26	3.2	49.9 × 108.3	8.9
Wall plate	26	3.2	48.1 × 108.3	9.1
Sync gear	2	20.0	72 × 71.8	1.1
Far end plate	1	16.0	178 × 164.2	1.1
Drive end plate	1	16.0	194 × 164.2	1.2
Motor plate	1	12.0	194 × 164.2	1.1
Top plate	1	8.0	178 × 115.2	1
Bottom plate	1	8.0	170 × 83.2	0.8
Motor cradle	1	12.0	114.8 × 120	0.6
Laser total	**110**			**41.7**

WHAT CHANGED IN THIS REVISION (COST CUTS)

- Base: 10 mm steel → 18 mm plywood. That was the biggest laser part (2 371 cm², 2.8 m of cut); total machine mass goes from 52.5 kg to 36.8 kg.
- Gear guard: 2 mm sheet metal → printed PETG, so no sheet-metal work. Gears: pre-hard 4140 bought as plate, so no heat treat or warping.
- Blades and plates: start in mild steel. 104 thin parts are 36 of the 41.7 m of laser cut, so ask for one nested quote rather than 11 separate ones.
- 26 blades, 26 spacers, the gears, the plates and every purchased part are unchanged. Thickness note: 3.0 / 3.2 / 3.4 mm are not stock sizes everywhere; tell me what the supplier carries and I will re-pitch the stack.

All dimensions in mm | Steel plate, laser / waterjet cut | Name: Ryan Boe

ASSEMBLY ORDER, DESIGN CHECKS AND OPEN ITEMS

ASSEMBLY ORDER

- Cut and deburr all plates; take the two hex bars to a machine shop to turn the ends to Ø25 h6 (far seat 17 long; shaft A drive end 15 + round to the coupler).
- Press 6205 bearings into the far and drive plates.
- Stack shaft A (blade first) and shaft B (spacer first) on their hex bars.
- Lay the far plate flat; drop in both shafts, then alternate comb / wall plates on the four rods.
- Fit the drive plate; thread the rods into it and snug evenly.
- Slide on the gears (set the tooth phase), then fit the gear guard.
- Bolt the gearbox to the motor plate from behind (4× M6 × 20), then the motor to the gearbox (remove its key, clamp the collar).
- Standoffs (M8 × 130) join the motor plate to the drive plate; slide the coupler on and clamp it 4.0 mm from the gear.
- Motor cradle, base, top / bottom plates, throat, lid + reed switch, chute.
- Wire the E-stop loop first; set 3.7 A; first spin with no load, then jog.

DESIGN CHECK	RESULT
Blade tip ↔ opposing spacer	1.08 mm
Blade ↔ stationary plate (axial)	0.10 mm
Blade tip ↔ wall plate	1.50 mm
Gear backlash at 68.0 centres	0.09 mm
Gear back face ↔ coupler	4.0 mm
Shaft engagement A / gearbox	29 / 28 mm of 30
Gear stress at the cap (hardened)	144 bend / 828 contact MPa
Bearing static factor (worst case)	× 1.7
Mesh-vs-mesh interference, 142 parts	none (rods thread into drive plate)

OPEN ITEMS – CHECK ON ARRIVAL

- Motor: 169 mm is treated as overall incl. the encoder (+28 mm if it is the body only); weight 5.5 kg is assumed.
- Gearbox: shaft length is taken from the pilot tip; ratings and the 235 g / 2.4 kg weight come from the listing – confirm the key and input clamp.
- Coupler: ask the seller whether the bores are keyed; shaft A end needs no key for the clamp type.
- Motor torque-vs-speed curve at 60 V is assumed (75 / 60 / 40 % of hold).
- Gears: pre-hard 4140 only (mild steel fails the contact check); bearing retention and hex tolerances are not drawn.
- Stack tolerance: 26 layers of stock vary by ±0.05 mm; measure the plate and re-sort parts so the blades do not rub the combs.
- Plate stock: 3.0 / 3.2 / 3.4 mm set; bit size and speeds are estimates – test with scrap PLA / PETG.